

Mehmet Akif Duran

Computer Engineering Student

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Computer Engineering student with a focus on artificial intelligence and data science, passionate about developing data-driven solutions and intelligent algorithms. I combine solid engineering principles with design thinking to build solutions that are both visually compelling and rigorously crafted for performance and usability.

EXPERIENCE

Social Media Specialist

Meet and Study Library September 2025 - August 2026

- Developed promotional strategies for library events across social media platforms.
- Created a content calendar to manage consistent and engaging post flow, increasing follower engagement rates.

Information Technologies Intern

4S Information Technologies August 2025 - September 2025

- Worked on Active Directory installation, administration, and maintenance.
- Played an active role in network infrastructure setup, configuration, and troubleshooting.
- Performed system monitoring and log analysis to identify potential security vulnerabilities.

Research & Development Intern

Feed Protocol October 2024 - January 2025

- Supported the development of blockchain technologies and decentralized applications (dApps).
- Conducted research on Zero Knowledge Proof (ZKP) technologies and authored technical reports.
- Conducted in-depth studies on Verifiable Random Functions (VRF) and Random Number Generation (RNG) mechanisms.

Human Resources Intern

Sports Center Academy February 2023 - May 2023

- Actively contributed to recruitment processes including CV screening and preliminary interview coordination.
- Assisted in organizing training and development programs while tracking employee participation.
- Supported administrative HR processes such as organizing and updating personnel files.

EDUCATION

B.Sc. Computer Engineering (30% English)

Başkent University September 2023 - January 2027 (Expected)

B.Sc. Computer Engineering (100% English) — Transfer

Istanbul Bilgi University 2022 - 2023

High School Diploma

Meram Anatolian High School 2017 - 2021

LANGUAGE PROFICIENCY

- YÖKDİL (English):** 78.75 / 100 — Turkish national foreign-language proficiency exam

TECHNICAL SKILLS

- Programming Languages:** C/C++, C#, Java, Python, MATLAB
- Systems & Network Programming:** Multi-threaded/socket-based applications in C/C++; TCP/IP and UDP protocols
- Embedded Systems:** System programming and embedded projects with Arduino and Proteus
- Software Development:** OOP, SOLID principles, Design Patterns, Algorithms and Data Structures
- Other:** Blockchain / dApps, Zero Knowledge Proof (ZKP), Active Directory, Network Administration

PROJECTS

MedAssist — Medication Tracking & Caregiver Management Mobile App

React Native · TypeScript · Node.js/Express · PostgreSQL · Redis/BullMQ · Stripe · FCM/APNs

- Built a cross-platform React Native (TypeScript) client with centralized state management and offline-first caching to track medication adherence for elderly and chronically ill patients.
- Designed a normalized PostgreSQL schema (patients, caregivers, prescriptions, dose events) with role-based access control linking each caregiver to multiple dependents.
- Engineered a Redis/BullMQ job queue for timezone-aware dose scheduling, dispatching push notifications via FCM/APNs with retry/backoff and missed-dose escalation to caregivers.

- Exposed a RESTful Node.js/Express API secured with JWT authentication and integrated Stripe (webhooks, idempotent payments) for subscription billing.

Başkent University Course Planning & Classroom Reservation System

Constraint Satisfaction (CSP) · Backtracking · Graph Coloring · Local Search

- Modeled university timetabling as a constraint-satisfaction problem, implementing backtracking with forward-checking and graph-coloring heuristics to eliminate instructor, room, and time-slot conflicts.
- Developed a greedy plus local-search optimizer to minimize schedule gaps and balance classroom utilization under capacity and equipment constraints.
- Built a reservation module with transactional conflict detection to prevent real-time double-booking of classrooms.

Stoxly — Investment Advisory Platform

REST/WebSocket · Time-Series (OHLCV) · Portfolio Analytics · Data Visualization

- Engineered a data pipeline ingesting real-time market feeds over REST/WebSocket APIs and persisting OHLCV time-series for downstream analysis.
- Implemented portfolio analytics computing risk/return metrics (Sharpe ratio, volatility, beta) and mean-variance optimization to generate asset-allocation recommendations.
- Delivered interactive dashboards visualizing performance attribution and rebalancing reports.

File Bridge — Secure File Transfer Application

SFTP/SSH2 · Multi-threading · End-to-End Encryption

- Developed an SFTP client over the SSH2 transport layer with public-key/password authentication and host-key verification for end-to-end encrypted transfers.
- Implemented chunked, resumable transfers with byte-offset tracking and integrity checks, backed by a multi-threaded transfer queue with connection pooling.
- Built a session manager for encrypted credential storage and a drag-and-drop UI with real-time progress and throughput reporting.

ECG Signal Analysis & Arrhythmia Detection

C++ · MATLAB · DSP · Pan-Tompkins

- Implemented a C++/MATLAB DSP pipeline performing bandpass filtering and baseline-wander removal, followed by Pan-Tompkins QRS detection for R-peak localization.
- Derived RR-interval and heart-rate-variability features and applied threshold-based classification to flag arrhythmias (bradycardia, tachycardia, irregular rhythm), validated against annotated ECG datasets.

Smart Trash Bin & Mobile Application

Arduino · Android (Java) · Ultrasonic Sensors · Servo Motors

- Built an Arduino-based embedded system using ultrasonic distance sensing to drive servo-actuated touchless lids across separate recyclable and non-recyclable compartments.
- Streamed real-time fill-level telemetry to a companion Android (Java) app rendering capacity gauges and recycling statistics.

Traffic Light Simulation System

Finite-State Machine (FSM) · Adaptive Timing · Simulation

- Designed a finite-state-machine controller coordinating multi-intersection signal phases with configurable green/yellow/red timing cycles.
- Implemented sensor-driven adaptive timing and a manual override mode to simulate congestion and edge-case traffic-flow scenarios.

VOLUNTEERING

Lead — Google Developer Groups on Campus

Başkent University *September 2025 – September 2026*

- Led the GDG student community at Başkent University, organizing technology events, workshops, and training sessions.

President — Başkent Computer Society

Başkent University *October 2024 – October 2025*

- Led the university Computer Society, organizing technical events and workshops.
- Fostered collaboration and knowledge sharing among computer engineering students.

REFERENCES

References available upon request.